

The Method of Monte Carlo

- How to generate observations from a specific distribution?
- This is an important topic. Bayesian, bootstrap.
- A generator of random uniform observation is all that is needed,

$$X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{unif}(0, 1)$$

- A discrete random variable:

$$X = \begin{cases} 1 & \text{if upface} \in \{1, 2\} \\ 0 & \text{o.w.} \end{cases}$$

$$\mathbb{P}(X = 1) = \frac{1}{3} \text{ and } \mathbb{P}(X = 0) = \frac{2}{3}$$

- If $U \sim \text{unif}(0, 1)$,

$$X = \begin{cases} 1 & \text{if } 0 < U < \frac{1}{3} \\ 0 & \text{if } \frac{1}{3} \leq U < 1 \end{cases}$$

- $\mathbb{P}(X = 1) = \mathbb{P}(0 < U < \frac{1}{3}) = \int_0^{1/3} dx = \frac{1}{3}$
 $\mathbb{P}(X = 0) = \mathbb{P}(\frac{1}{3} < U < 1) = \int_{1/3}^1 dx = \frac{2}{3}$

U_i	0.4743	0.7891	0.5550	0.9693	0.0299	0.8425	0.6012	0.1009	0.0545	0.4679
X_i	0	0	0	0	1	0	0	1	1	0

- If $(U_1, U_2) \stackrel{i.i.d.}{\sim} \text{unif}[0, 1]^2$, the probability of (U_1, U_2) lying within the unit circle is $\frac{\pi}{4}$.

- Define

$$X = \begin{cases} 1 & \text{if } U_1^2 + U_2^2 < 1 \\ 0 & \text{o.w.} \end{cases}$$

$$\mathbb{E}(X) = \mathbb{P}(U_1^2 + U_2^2 < 1) = \int \int_{u_1^2 + u_2^2 < 1} du_1 du_2 = \frac{\pi}{4}$$

- Observe $X_1, \dots, X_n$. Since $\mathbb{E}X = \pi/4$, $4\bar{X}$ is an unbiased estimation of π

n	100	500	1000	10,000	100,000
$4\bar{X}$	3.24	3.072	3.132	3.138	3.13828

- Let $U \sim \text{Unif}(0, 1)$. Let F be a continuous cdf. Then, $X = F^{-1}(U)$ has the distribution function F .

$$\begin{aligned}\text{proof: } \mathbb{P}(X \leq x) &= \mathbb{P}(F^{-1}(U) \leq x) \\ &= \mathbb{P}(U \leq F(x)) \\ &= \int_0^{F(x)} dx \\ &= F(x)\end{aligned}$$

- Let $X \sim \exp(\lambda)$ with pdf $f(x) = \lambda e^{-\lambda x}$ for $x > 0$, and cdf $F(x) = 1 - e^{-\lambda x}$ for $x > 0$.
- $F^{-1}(x) = -\frac{1}{\lambda} \log(1 - x)$. Define

$$X = -\frac{1}{\lambda} \log(1 - U) \sim \exp(\lambda), \quad U \sim \text{unif}(0, 1)$$

u	0.473	0.858	0.501	0.676	0.240
x	0.641	1.950	0.696	1.130	0.274

- Objective: $\int_a^b g(x)dx$
- Note that

$$\int_a^b g(x)dx = (b-a) \int_a^b g(x) \frac{1}{b-a} dx = (b-a)\mathbb{E}[g(X)], \quad X \sim \text{unif}(a, b).$$

- $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{unif}(a, b)$. Let $Y_i = (b-a)g(X_i)$. Then $\bar{Y}$ is an unbiased estimator of $\int_a^b g(x)dx$.

- Estimation of π
- Let $g(x) = 4\sqrt{1-x^2}$, $0 < x < 1$.

$$\pi = \int_0^1 g(x)dx = \mathbb{E}[g(X)], \quad X \sim \text{unif}(0, 1)$$

- $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{unif}(0, 1)$. $Y_i = 4\sqrt{1-X_i^2}$. Then $\bar{Y}$ is an unbiased estimator of π .

n	100	1,000	10,000	100,000
$\bar{Y}$	3.2718	3.1033	3.1454	3.142066

- Let $(Y_1, Y_2) \sim \text{unif}(0, 1)^2$. Define

$$X_1 = (-2 \log Y_1)^{1/2} \cos(2\pi Y_2)$$

$$X_2 = (-2 \log Y_1)^{1/2} \sin(2\pi Y_2)$$

- one-to-one transformation: $\{(y_1, y_2) : 0 < y_1, y_2 < 1\}$,
 $\{(x_1, x_2) : -\infty < x_1, x_2 < \infty\}$. The inverse transformation is

$$y_1 = \exp\left(-\frac{x_1^2 + x_2^2}{2}\right)$$

$$y_2 = \frac{1}{2\pi} \arctan \frac{x_2}{x_1}$$

- Jacobian:

$$J = \begin{vmatrix} -x_1 e^{-\frac{x_1^2 + x_2^2}{2}} & -x_2 e^{-\frac{x_1^2 + x_2^2}{2}} \\ -\frac{x_2/x_1}{2\pi(1+x_2^2/x_1^2)} & \frac{1/x_1}{2\pi(1+x_2^2/x_1^2)} \end{vmatrix} = \frac{1}{2\pi} e^{\frac{x_1^2 + x_2^2}{2}}$$

- The joint pdf of X_1 and X_2 is

$$f(x_1, x_2) = \frac{1}{2\pi} e^{\frac{x_1^2 + x_2^2}{2}}$$

So, X_1 and X_2 follow $N(0, 1)$ independently.

- Let f be a pdf. Suppose that Y is a random variable with pdf $g(y)$, $U \sim \text{unif}(0, 1)$, and Y and U are independent. We also have

$$f(x) \leq Mg(x), \quad -\infty < x < \infty.$$

The following algorithm generates a random variable X with pdf $f(x)$.

1. Generate Y and U .
2. If $U \leq \frac{f(Y)}{Mg(Y)}$, then take $X = Y$. Otherwise return to step 1.
3. X has pdf $f(x)$.

- Suppose $X \sim N(0, 1)$ and Y has a Cauchy distribution with pdf $g(y) = \frac{1}{\pi(1+y^2)}$.
- Ignoring normalizing constants, the ratio to bound is

$$\frac{f(x)}{g(x)} \propto (1 + x^2) \exp(-x^2/2) \leq 2 \exp(-1/2) = 1.213.$$

So, $M = (\pi/\sqrt{2\pi})1.213 = 1.520$. Hence, the acceptance rate of the algorithm is $1/M = 0.6577$.

References

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- [3] *A First Course in Probability* by Ross
- [4] *John E. Freund's Mathematical Statistics with Applications* by Irwin Miller Marylees Miller